Table 62. IWDG min/max timeout period at 32 kHz (LSI)

e: *For the values in this table, the exact timings still depend on the phasing of the APB interface clock versus the LSI clock, so that there is always a full RC period of uncertainty.*

Prescaler divider	PR[2:0] bits	Min timeout RL[11:0] = 0x000	Max timeout RL[11:0] = 0xFFF	Unit
/4	0	0.125	512	
/8	1	0.250	1024	
/16	2	0.500	2048	
/32	3	1.0	4096	ms
/64	4	2.0	8192	
/128	5	4.0	16384	
/256	6 or 7	8.0	32768	

Table 63. WWDG min/max timeout value at 144 MHz (PCLK)

Prescaler	WDGTB	Min timeout values	Max timeout value	Unit
1	0	0.0284	1.820	
2	1	0.0569	3.641	
4	2	0.1138	7.282	
8	3	0.2276	14.564	
16	4	0.4551	29.127	ms
32	5	0.9102	58.254	
64	6	1.820	116.508	
128	7	3.641	233.017	

5.3.22 I3C interface characteristics

The I3C interface meets the timing requirements of the MIPI® I3C specification v1.1.

The I3C peripheral supports:

- I3C SDR-only as controller
- I3C SDR-only as target
- I3C SCL bus clock frequency up to 12.5 MHz

The parameters given in Table 64 below are obtained with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- I/O Compensation cell activated
- Voltage scaling range 1

The I3C timings are in line with the MIPI specification, except for the ones given in Table 64, I3C open-drain measured timing. For t_{SU_OD}, this can be mitigated by increasing the corresponding SCL low duration in the I3C_TIMINGR0 register. For further details refer to AN5879.

Table 64. Open drain timing measurements

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions	Min	Unit
t_{SU_OD}	SDA data setup time in open drain mode	Controller 2.7 V ≤ V_{DDIOX} ≤ 3.6 V	22(1)	ns

1. *The minimum SDA data setup time during open-drain-mode is 3 ns, as specified in the MIPI Alliance specification for I3C.*